# LedgerForge - Phase 3 Submission Report

Sudharsana | Python Track | LedgerForge | Phase 3 - Connected System

Repository: github.com/Sudharsana27/LedgerForge | Git tag: phase-3 (retained) | 7 November 2026

## Executive summary

Phase 3 extends the existing FastAPI and SQLAlchemy ledger with authenticated access, auditable payroll and expense workflows, versioned tax configuration, and financial statements. Earlier employee, salary, attendance, leave, payroll-calculation, payslip, invoicing, and double-entry ledger behavior remains covered by regression tests.

## Five-task completion status

| Task | Status | Evidence / limitation |
|---|---|---|
| Partial component audit | 7339/7339 fixed matches; 7337/7339 overtime matches; 44 ESI conflicts; one attendance row unavailable. Not a complete Check 2 pass. | |
| 11 - Expenses | Completed and independently verified | 501 bills, 18 months; monthly and vendor-month totals match; workflow tests pass. |
| 12 - Payroll | Partially completed | Lifecycle, idempotency, posting and reversal tested; complete Check 2 remains blocked; separate partial component audit generated. |
| 13 - Security | Implemented but not independently verified | 64-byte local key; 30-minute login and route tested; source users have no passwords. |
| 14 - Tax | Partially completed | Audited effective versions; 54/54 monthly sums match reference records; policy is sample data. |
| 15 - Reports | Completed and independently verified | 5/5 official checks; trial balance 40 accounts, 0 mismatches; balance sheet balances. |

## Authentication and authorization

API access uses 30-minute signed bearer tokens. Password hashes are verified with bcrypt, user activation is checked on every request, and employee-facing routes enforce record ownership. ADMIN and HR roles protect operational and financial surfaces; only ADMIN may version tax configuration. Cross-employee payslip access is HR-only. A generated 64-byte secret is stored only in the ignored local .env. Login with that key returned 200, a 1,800-second token, and 200 on a protected route. Automated tests cover missing/weak keys, 401/403, role isolation, and inactive-user revocation. Source users have no passwords.

## Payroll lifecycle

Runs transition DRAFT -> PROCESSED -> APPROVED -> PAID. Processing recalculates saved items through the earlier payroll engine; approval links its balanced journal atomically and enforces separation of duties. Repeated approval, payment, and reversal do not duplicate postings. Reversal entries offset both payroll and recorded payment journals.

# Financial, expense, and tax controls

## Expense workflow

Expense bills progress through SUBMITTED, APPROVED, and PAID. The workflow prevents self-approval, detects duplicate vendor / reference / date / amount combinations, validates posting accounts, and records balanced accrual and settlement journals. Spend reconciliation compares monthly and vendor-month totals against the supplied expenses.csv source.

## Financial reporting

Trial balance, profit-and-loss, and balance-sheet reports include posted journal activity only. Contra-asset normal balances and unclosed current-period profit are represented so the balance sheet remains balanced. The trial-balance export is formatted for comparison by the official LedgerForge checker.

## Tax and statutory reporting

Tax slab changes create new effective-dated versions and audit records instead of rewriting previously saved payroll. Admin routes now version tax parameters and statutory rates with audit history. TDS uses configured slabs, deduction, rebate threshold, and cess; configuration changes and restoration are tested. PF, ESI, and TDS monthly report totals match the stored reference payroll items in 54/54 month/type checks. This checks report summation, not legal correctness: source rules are explicitly training samples.

## Source-data constraints

The supplied data has 18 payroll runs and 7,340 items but no independent payroll_inputs.csv. The independent batch validator processed 7,340 eligible employee/run rows and skipped all 7,340 for the same reason: missing independent variable-pay, TDS, and other-deduction inputs. The official Check 2 found 0/7,340 result rows; no payroll match is claimed. A separate source-only partial audit reconstructs fixed components for 7339/7340 employee/run rows and attendance-dependent values for 7339 rows; 1 row is unavailable because attendance is missing.

**PARTIAL RECONCILIATION — NOT A COMPLETE CHECK 2 PASS.** Overtime pay matches 7337/7339 calculable rows; the 2 explicit HALF_UP/reference one-paisa differences are EMP-0307/PR-2025-11 4898.63 vs 4898.62, EMP-0535/PR-2026-07 6361.13 vs 6361.12. PF employee/employer matches are 7340/7340 and 7340/7340; these use sample rates. 44 ESI rows remain unresolved source-data conflicts. No variable pay, TDS, other deductions, complete gross, or net pay is claimed.

Tax and statutory rate rows are explicitly marked sample or training data. The existing source also documents conflicting ESI eligibility and two one-paisa overtime rounding differences. The calculation engine preserves its explicit input and rate rules; it does not alter historical values to force a match.

**Permissions:** Admin manages operations and tax versions; HR has organization-wide HR/finance/report access; employees access their own records; public access is limited to health and login. Admin tax changes are tested as 201; HR/employee changes as 403.

# Validation and release record

## Automated and official checks

| Validation | Result |
|---|---|
| Pytest regression + Phase 3 | 181 passed; 1 warning |
| Independent payroll Check 2 | 0/7,340 found; 7,340 skipped for missing independent inputs; not a pass |
| Monthly statutory report sums | PF, ESI, TDS: 54/54 matches to source-item totals; not statutory policy validation |
| Official source checks | 5/5 passed |
| Official trial balance | 40 accounts; 0 mismatches |
| Expense reconciliation | 501 bills, 18 months; 0 monthly or vendor-month mismatches |
| Migration | Backed up; upgraded a390f1b6d2e8 → d1e4a6b8c0f2; source row counts preserved; integrity OK |

## Official vs. LedgerForge totals

| Measure | Official | LedgerForge |
|---|---|---|
| Revenue | 1,260,930,052.00 | 1,260,930,052.00 |
| Expenses | 1,036,246,464.42 | 1,036,246,464.42 |
| Profit | 224,683,587.58 | 224,683,587.58 |

Official payroll summary: 18 runs; employer cost 861,586,392.42. Balance sheet at 2026-09-30: assets 463,762,958.65; liabilities 77,499,607.07; equity 161,579,764.00; current profit 224,683,587.58; difference 0.00.

## Operational notes

The original phase-3 tag is unchanged. The additive audit migration is d1e4a6b8c0f2; a fresh SQLite upgrade and the backed up local database both reached head. Existing payroll, journal, and tax source row counts were preserved; integrity_check=ok.

Before enabling login, configure a strong local SECRET_KEY and provision active users with the interactive password utility. Before treating tax or historical payroll as production-ready, obtain approved statutory policy and independent payroll input records. All supplied company data is fictional training data.

## Problems and resolutions

Payroll inputs are incomplete, so the 7,340 Check 2 skips are reported rather than filled with reference values. Tax and statutory sources are sample-only; report matches are limited to stored totals. A verified backup preceded the audit migration. A weak local JWT key was rotated without printing or committing the replacement.

## Learning and Phase 4

I learned to separate authorization from employee ownership, and to make journal operations atomic and idempotent.
I learned that effective-dated settings need append-only audit history and must preserve saved payroll.
Reconciliation must use independent inputs; reference outputs cannot stand in for missing calculation inputs.
Phase 4: obtain approved payroll/tax sources, then harden deployment, monitoring, backups, and recovery.